

18 Steady state stability

Why Are We Learning This?

In previous energy-balance lectures, we learned that a non-isothermal reactor must simultaneously satisfy: Mole Balance and Energy Balance.

For some exothermic reactions, these two balances can be satisfied at **more than one reactor temperature**. As a result, the same reactor operating under the same conditions may have multiple possible steady states. This phenomenon is one of the most important—and potentially dangerous—features of non-isothermal reactor design. Small changes in feed temperature can cause:

Large Changes in Reactor Temperature

Large Changes in Conversion

Large Changes in Heat Generation

Understanding ignition, extinction, and stability is essential for designing reactors that operate safely and predictably.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
► One Reaction: Non-Isothermal Reactor Design ◀
↓
Multiple Reactions
↓
Reactor Safety
↓
Challenging the Well-Mixed Assumption
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Learning Objectives

By the end of this lesson, you should be able to:

- Explain how multiple steady states arise in non-isothermal CSTRs.
- Interpret heat-generated and heat-removed curves.
- Determine the number of possible steady states from a $G(T)$ – $R(T)$ plot.
- Distinguish between stable and unstable steady states.
- Predict how a reactor responds to small temperature perturbations.
- Explain the concept of local stability.

Big Idea

Heat Generated

vs

Heat Removed

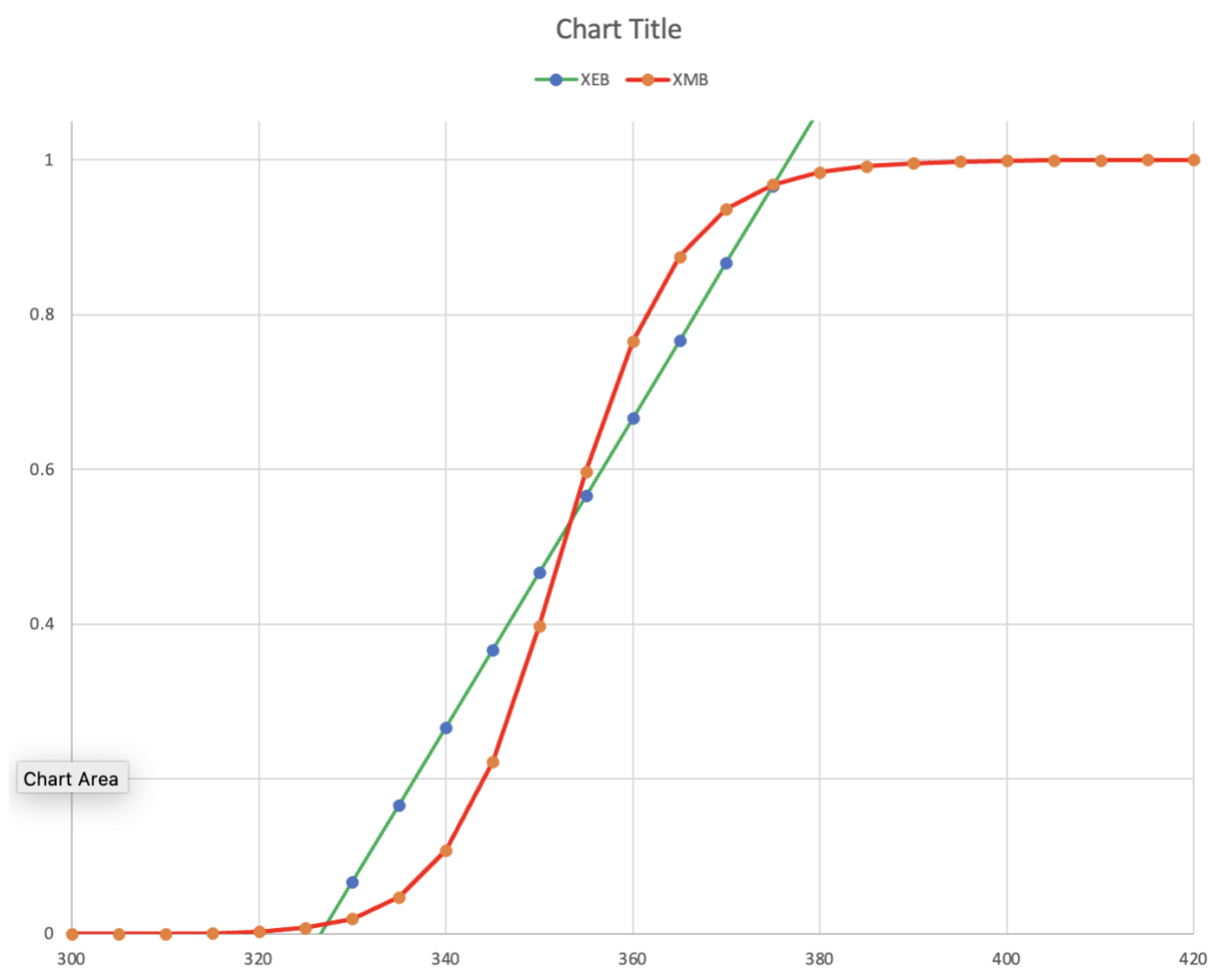


Determines Steady States



Determines Reactor Stability

A reactor is stable only if small disturbances cause it to return to its original operating condition.



- The steady state temperatures are the temperatures where

Why Do Intersections Matter?

At steady state: Heat Generated = Heat Removed

If: $G(T) > R(T)$ the reactor gains energy and the temperature increases.

If: $R(T) > G(T)$ the reactor loses energy and the temperature decreases.

Therefore: $G(T) = R(T)$ defines a possible steady-state temperature.

Physical Interpretation

Heat Generated = Heat Removed



Temperature Remains Constant

Each intersection corresponds to a reactor temperature that simultaneously satisfies both the mole balance and the energy balance.

- The temperatures on the x-axis are
- There can be multiple points of intersection for one feed temperature. For example, the feed temperature above has 3 steady state solutions.

One Feed Temperature Can Have Multiple Steady States

Students often assume:

One Feed Temperature



One Reactor Temperature

However, this is not always true. For some exothermic reactors:

One Feed Temperature



Several Steady States

may satisfy both the mole balance and the energy balance simultaneously.

Physical Interpretation

Heat Generated Curve intersects Heat Removed Curve multiple times; Each intersection represents a possible steady-state reactor temperature.

- Steady state stability depends on

- If $G(T) > R(T)$

- If $R(T) > G(T)$

How Do We Determine Stability?

A steady state is stable if a small disturbance causes the reactor to return to that steady state. A steady state is unstable if a small disturbance causes the reactor to move away from that steady state.

Stability Test

Imagine increasing the reactor temperature slightly.

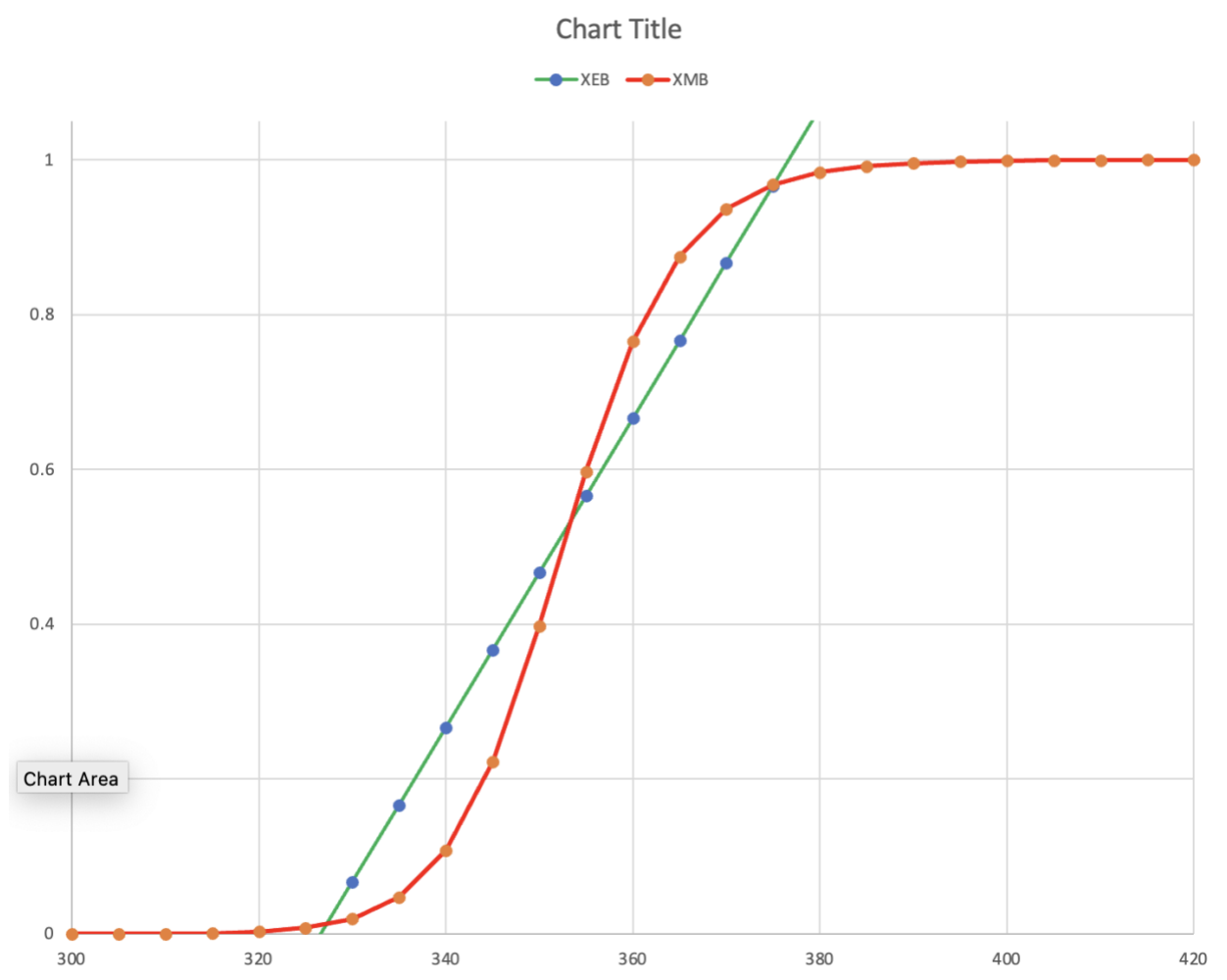
If: $R > G$ the reactor cools back down. Good. Stable.

If: $G > R$ the reactor heats up further. Bad. Unstable.

Key Question

After a Disturbance Does the Reactor Return or Move Away? That determines stability.

- Hence, in the plot above:
 - Reactor temperature = 324 K:
 - Reactor temperature = 354 K:
 - Reactor temperature = 380 K:



Stability Decision Guide

Is 324 K Stable?

Ask: If Temperature Rises Slightly, Does the Reactor Return?

If the answer is yes: Stable

If the answer is no: Unstable

Is 354 K Stable?

Check both directions: Small Temperature Increase and Small Temperature Decrease

If both disturbances move the reactor away from 354 K: 354 K is Unstable

Is 380 K Stable?

Ask: If Temperature Falls Slightly, Does the Reactor Return?

If the answer is yes: Stable

If the answer is no: Unstable

Why Is the Middle Steady State Unstable?

This is the most important stability result. Suppose the reactor is operating at the middle steady state. If the temperature increases slightly: $G > R$ and temperature rises further.

If the temperature decreases slightly: $R > G$ and temperature falls further.

Physical Interpretation

Small Disturbance



Moves Away From Steady State.

The reactor does not recover. Instead, it moves toward one of the neighboring stable states.

Key Result

Middle Branch = Unstable in the classic ignition-extinction curve.

Why Are the Upper and Lower Branches Stable?

Consider a stable steady state. If temperature rises slightly: $R > G$ so the reactor cools back down.

If temperature falls slightly: $G > R$ so the reactor heats back up.

Physical Interpretation

Small Disturbance



Forces Reactor Back Toward Steady State

Stable operating points behave like a ball sitting in a valley:

Push It Slightly



Rolls Back

Unstable operating points behave like a ball balanced on top of a hill:

Push It Slightly



Rolls Away

Summary and Key Takeaways

- ✓ Steady states occur where: $G(T) = R(T)$
- ✓ One feed temperature can produce multiple steady states.
- ✓ Stability depends on how the reactor responds to small temperature disturbances.
- ✓ Lower and upper steady states are usually stable.
- ✓ Middle steady states are usually unstable.

Most Important Idea

Steady State



Perturb Temperature



Observe Direction of Motion



Determine Stability

A reactor operating point is not important simply because it satisfies the mole and energy balances. To be observed in practice, the operating point must also

be stable—that is, the reactor must return to that state after a small disturbance.